

Information Propagation Methods for Reliability, Reachability and Flow Analysis in Directed Acyclic Process Networks

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Viva voce, 9 October 2026

The problem

An engineer asks three questions of one system, such as a water network:

- Will supply reach each node? *reliability*
- How much can it deliver? *capacity*
- How long to restore it? *schedule*

Today each is a separate model, so the **topology** is **entered three times** and kept consistent by hand.

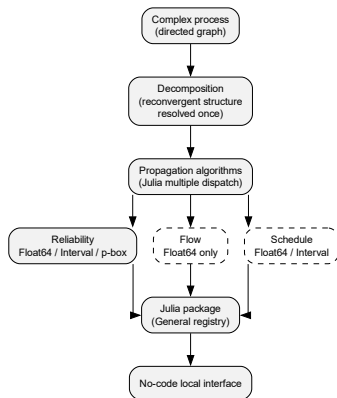
Every input is uncertain, but it is **collapsed to a number** before any analysis runs.

Why it is hard

Redundancy is *reconvergence*: routes split and rejoin, sharing components upstream. The independence assumption of simple propagation then fails, and always **overstates** reliability.

No reviewed tool combines all three analyses on one network, and none propagates intervals or p-boxes natively.

One model, three interpretations

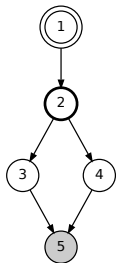


A system is a **DAG**. One **graph object** (layers, ancestor and descendant closures, node roles) is read by three toolkits.

Research questions

- 1 Can one DAG model serve all three analyses, topology entered once?
- 2 Can reconvergence be identified once and reused?
- 3 Exact reliability under intervals and p-boxes: what is guaranteed, at what cost?
- 4 Which critical-path quantities stay exact under interval durations?
- 5 Can it reach an engineer who does not program, with the model kept on their machine?

Reconvergence, identified once and conditioned on



every component 0.9

Independent update at join 5: $b(5) = 0.749$. Exact: $b(5) = 0.675$.

Condition on the fork's reachability, then recombine by total probability:

$$b(5) = 0.9 \left[\underbrace{0.81 \times (1 - (1 - 0.729)^2)}_{\text{given fork reached}} + 0.19 \times 0 \right] = 0.675462$$

Two sub-problems replace 2^{10} component states.

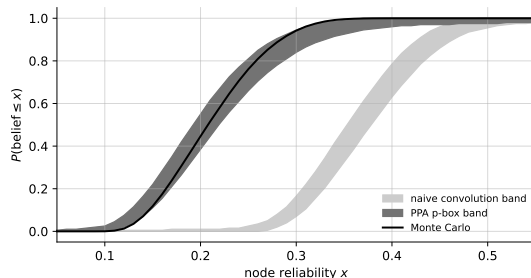
Network Decomposition Module: finds every *diamond* (a fork whose paths reconverge at a join), stores each unique diamond once with its **conditioning set**, and is proved complete, deterministic and terminating.

Probability Propagation Toolkit: exact, and what it costs

One pass over the layers, conditioning at each join.

- **Point inputs:** exact belief; agrees with a sifted BDD to 1.1×10^{-16} on 129 graphs and named families.
- **Intervals:** exact range, because belief is monotone in every input.
- **p-boxes:** sound bounds on the distribution of each belief.

Cost: exponential in the largest conditioning set, the same width that governs a well-ordered decision diagram. Neither method dominates; the case rests on native interval and p-box propagation.



p-box band brackets the Monte Carlo distribution

Capacity and schedule on the same graph object

Capacity Flow Toolkit (point-valued)

- Throughput, minimum cuts, saturated edges, single points of failure, degradation and upgrade thresholds, redundancy.
- Every minimum cut from one solve: the closed subsets of the free zone (Picard and Queyranne), at most $2^{|\mathcal{F}|}$.
- Two designs, same $F^* = 37$: one has 4 minimum cuts, the other 1. Throughput alone does not tell them apart.

Critical Path Toolkit

- CPM generalised to a family of operator pairs.
- Interval durations: forward quantities exact from two corner runs.
- Floats are NP-hard in general. The **domination split** enumerates only the bypass set H_v : cost $2 \sum_v 2^{|H_v|}$ against 2^k .
- PSPLIB: 50,524 propagations against 2^{30} ; three necessarily critical activities the enclosure misses.

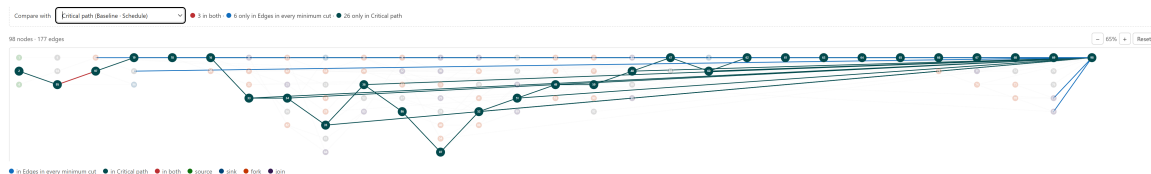
End to end: EPANET Net3

97 nodes, 119 edges, oriented by its own hydraulic simulation; converted once, three toolkits.

- **Reliability:** worst-served junction 0.81; [0.35, 0.99] under $\pm 5\%$.
- **Capacity:** whole demand met, 680.1 L/s; river pump out, 43.3 L/s (6.4%).
- **Schedule:** 74 days along 28 activities; [59, 89] days under intervals.

The **river pump** (95–97, red) is in every degraded minimum cut *and* on the restoration critical chain, a link only the two analyses together show.

Corrected since submission: the super-sink edges were missing from the flow graph. Errata sheet provided.



Degraded minimum cut (blue) against the baseline critical path (dark): 3 nodes in both, 95, 97 and the super-sink.

Answers, contributions and limits

Answers

- 1 Yes, on the model's four assumptions.
- 2 Yes, for the probabilistic interpretation.
- 3 Exact for points and intervals, sound for p-boxes; cost set by conditioning width.
- 4 Forward quantities in two runs; floats exact at a cost set by bypass width.
- 5 Yes: registered Julia package and a local no-code interface.

Limits, stated in the thesis

- Practical to about fifty nodes of moderate reconvergence; p-boxes on small networks.
- Flow toolkit is point-valued.
- Acyclicity is a property of one operating snapshot.
- The interface is demonstrated, not yet studied in use.

Package:

`InformationPropagationAnalysis.jl`

Demo:

`information-propagation-framework.dev`

Thank you

Questions welcome

Backup: corrections since submission

Chapter 6. “Exactly $2^{|\mathcal{F}|}$ minimum cuts” becomes “at most”: $S^* \cup R$ is a minimum cut iff $R \subseteq \mathcal{F}$ is closed under residual reachability (Picard and Queyranne, 1980). Implementation corrected; every Chapter 6 result re-run and unchanged (Configuration A: 4 cuts; B: 1).

Chapter 10. The flow run used the edge list without the super-sink and its 58 demand edges, and reported 1,837.9 L/s, above the total demand of 680.1.

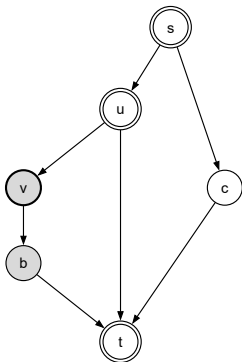
	Submitted	Corrected
Baseline throughput	1,837.9 L/s	680.1 L/s (all demand)
Baseline minimum cut	4 supply edges	the 58 demand edges
Degraded throughput	954.6 L/s	43.3 L/s (6.4%)
Degraded minimum cuts	same 4	24; river pump in every one
Overlay with critical path	“no node in common”	river pump on both

Checked against an independent Edmonds–Karp computation. Reliability and schedule unchanged.

Backup: why the degraded flow is so low

- With the river pump out, the sources are the lake pump and one discharging tank.
- In the snapshot orientation they reach only six junctions (7, 8 and 77–81), which are fully served; the other 52 receive less than their demand.
- The derated trunk main does not bind: its only feed is the river pump.
- Physically, the lake pump could push water through pipes that ran the other way at the snapshot. The oriented model cannot represent that. A degraded state needs an orientation of its own.

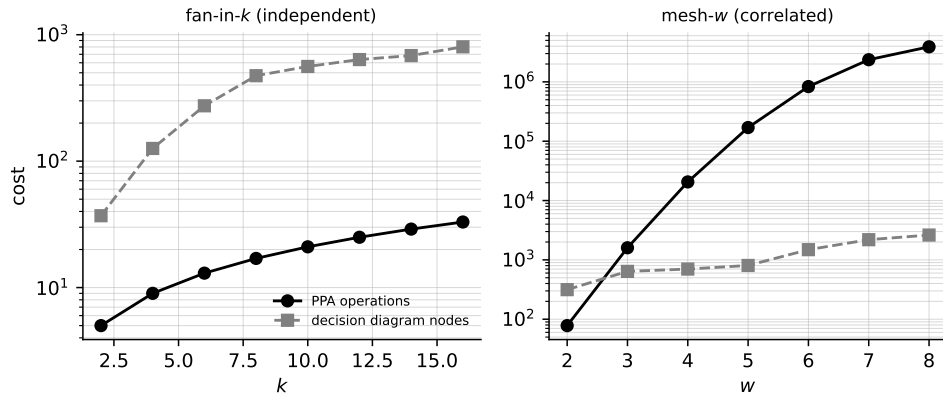
Backup: the domination split



Relative to node v : *dominated* durations (every route through them passes v) are fixed at one bound, *incomparable* ones at the other, and only the *bypass* set H_v , which shares routes with v and can route around it, is enumerated.

The exact floats then cost $2 \sum_v 2^{|H_v|}$ propagations in all, against 2^k for k interval durations.

Backup: cost against decision diagrams



Sub-problem count against sifted BDD node count on the two adversarial families (fan-in- k , mesh- w).